Phase 1 Report: Data Exploration and Setup

Initial Topic : Impact of Demographic, Clinical, and Structural MRI Indices on Dementia Severity

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1 Data Origin and Collection Mechanism

Official Source: The dataset is officially maintained and provided by the National Alzheimer's Coordinating Center (NACC), established and funded by the National Institute on Aging (NIA/NIH).

Collection Period: Data collection for the Uniform Data Set (UDS) was initiated in 2005 and continues as an ongoing, longitudinal data repository.

General Method and Study Type: This data originates from a **multicenter, longitudinal observational study** based on **clinical visits** and **medical imaging**. The data is systematically generated through standardized clinical evaluations, detailed neuropsychological assessments, and structural Magnetic Resonance Imaging (MRI) scans conducted by expert clinicians at various NIA-designated Alzheimer's Disease Research Centers (ADRCs) across the United States.

Access Limitations: The dataset is not open-access or publicly available. To protect patient privacy and ensure ethical use, access requires researcher registration, institutional affiliation, and the submission of a formal Data Use Agreement (DUA) and research proposal, which must be reviewed and approved by NACC.

2 Unit of Observation and Data Structure

Unit of Observation (Row-Level): Based on our initial data exploration, each row in the `udslong_imputed.csv` dataset represents a **single unique patient** at a specific point in time (most likely their baseline clinical evaluation).

Cross-Sectional Structure: Although the parent study (NACC UDS) collects longitudinal data, our exploratory analysis confirmed that every participant ID (`NACCID`) in this specific working dataset appears exactly once (0 patients with repeated visits). Therefore, for the analytical scope of this project, the data structure is **cross-sectional**.

Unlike a hierarchical or nested structure where multiple visits are grouped under a single patient, this dataset provides a single, independent observation per individual.

3 Sampling, Selection, and Generalizability

Sampling Mechanism: The individuals in this dataset were not selected through random population sampling. Instead, the cohort was assembled using **non-probability, clinic-based sampling** (primarily convenience and purposive sampling). Participants consist of individuals who either volunteered for research or were clinically referred to specialized Alzheimer's Disease Research Centers (ADRCs) for cognitive evaluation.

Study Design: The data is derived from an **observational study**. Researchers systematically record clinical histories, conduct neuropsychological tests, and acquire MRI scans without administering any experimental treatments or interventions.

Generalizability (External Validity): Due to the non-random sampling method, the dataset is inherently subject to selection bias (e.g., participants in such studies often have higher education levels or better healthcare access than the average citizen). Therefore, the findings of this study **can only be generalized to similar clinical populations** (i.e., patients attending specialized memory clinics). The results **cannot be generalized to the broader general population**, and they should not be used to estimate national disease prevalence.

4 Variable Classification and Measurement Scales

Table 1 outlines the classification of the selected variables based on their data type and measurement scale.

Table 1: Academic Classification of Variables by Type and Measurement Scale

Variable(s)	**Statistical Data Type**	**Measurement Scale**
NACCID	Identifier	Nominal
NACCAGE, NACCBMI	Quantitative	Continuous (Ratio)
LH/RH_ENTORHINAL_GVOL & AVGTH	Quantitative	Continuous (Ratio)
LH/RH_PARAHIPPOCAMPAL_GVOL & AVGTH	Quantitative	Continuous (Ratio)
EDUC, SMOKYRS	Quantitative	Discrete (Ratio)
NACCGDS	Quantitative	Discrete (Interval)
CDRSUM (Primary Response)	Quantitative / Categorical	Discrete / Ordinal
CDRGLOB, NACCUDSD	Categorical	Ordinal
NACCAPOE, DIABTYPE	Categorical	Nominal
SEX, TOBAC100, HYPERTEN, DIABETES, CBSTROKE	Categorical	Nominal (Binary/Dichotomous)

Methodological Note on Operational Treatment: While Table 1 reflects the rigorous statistical definitions, certain variables will be treated differently during the computational modeling phase (as outlined in the project's foundational documentation):

- **CDRSUM (Clinical Dementia Rating Sum of Boxes):** Strictly speaking, CDRSUM is a discrete/ordinal variable scoring from 0 to 18 in 0.5 increments. However, in accordance with standard practices in Alzheimer's research and neuroinformatics, it possesses sufficient granularity to be treated operationally as a *Continuous Quantitative* variable for continuous regression modeling.

5 Data Quality Audit and Handling Strategies

To ensure the integrity of the statistical analysis and predictive modeling, a comprehensive data quality audit was conducted. The strategies for identifying and handling various data anomalies are outlined below, adhering to best practices in clinical data science.

- **Missing Data:** *Strategy:* Features with a high percentage of missingness ($> 30\%$) will be excluded. For essential clinical variables with minor missingness, median imputation (for continuous) or mode imputation (for categorical) will be applied. Rows with missing target variables (`CDRSUM`) will be dropped. *Justification:* Dropping records with missing target variables prevents model hallucination, while imputing predictors preserves statistical power in the clinical cohort. (Note: As the current dataset is labeled `imputed.csv`, preliminary checks indicate minimal missingness).
- **Special Codes (Unknown / Not Answered):** *Strategy:* The NACC dataset utilizes specific numeric codes (e.g., 88, 99, 8, 9, -4) to represent "Unknown", "Not Applicable", or "Missing". These will be explicitly mapped to `NaN` (Not a Number) before any mathematical operations. *Justification:* If left unhandled, numeric missing codes (like an age of 999) will severely skew descriptive statistics, bias regression coefficients, and invalidate correlation metrics.
- **Impossible Values:** *Strategy:* Variables will be bounded by logical clinical limits (e.g., `NACCAGE` > 0, `NACCBMI` between 10 and 100). Furthermore, for categorical and binary variables such as `HYPERTEN`, `DIABETES`, and `CBSTROKE`, decimal values (e.g., 0.5 or 1.2) are fundamentally unacceptable. Any values violating these clinical boundaries or structural data types will be treated as data entry errors and converted to `NaN`. *Justification:* Non-physiological values and invalid structural types (such as decimals in nominal categories) compromise data integrity and disproportionately bias statistical modeling.
- **Duplicate Records:** *Strategy:* Exact duplicate rows across all columns will be dropped, retaining only the first occurrence. Furthermore, as evaluated in previous steps, duplicate `NACCID` entries will be flattened if the study design requires a strictly cross-sectional baseline cohort. *Justification:* Duplicate records artificially inflate the sample size, violating the assumption of independent observations and leading to underestimated standard errors.
- **Logical Inconsistencies:** *Strategy:* Relational integrity checks will be performed (e.g., verifying that Years of Education (`EDUC`) is not strictly greater than `AGE`). Flagged records will be evaluated and either corrected if the entry error is obvious, or removed. *Justification:* Logical discrepancies indicate compromised data provenance. Removing them prevents contradictory information from degrading the model's structural logic.

6 Empirical Distributions and Frequency Analysis

To understand the baseline characteristics of the study cohort, empirical distributions were analyzed. Frequency and percentage tables are provided for categorical variables, while descriptive statistical summaries are presented for quantitative measurements.

6.1 Categorical Variables: Frequencies and Percentages

Table 2 summarizes key demographic, clinical, and vascular risk categorical variables based on the cleaned dataset, explicitly indicating missing values (NaN).

6.2 Quantitative Variables: Distribution Summaries

Table 3 provides the central tendency and dispersion metrics for continuous and discrete quantitative variables extracted from both the clinical and structural MRI files after bounds enforcement and data cleaning.

6.3 Important Clinical Patterns (Post-Cleaning)

Beyond the raw tabular data, several critical patterns emerge from the empirical distribution of the cleaned dataset, dictating the direction of subsequent statistical modeling:

- **Confirmation of Successful Data Cleaning:** The distribution tables demonstrate that impossible values and missing codes have been successfully managed. For example, the maximum value for `NACCBMI` is now strictly bounded to a physiological 52.4, the maximum for `EDUC` is 25.0, and `NACCGDS` is correctly capped at 14.0. Additionally, invalid decimal codes in categorical variables have been removed and accurately classified as missing (NaN), such as 2.67% missing data for `HYPERTEN` and 1.62% for `DIABETES`.
- **Severe Target Variable Imbalance (Zero-Inflation):** The distribution of cognitive status exhibits extreme right-skewness. Exactly 64.67% of the cohort presents a completely normal cognitive status (`CDRGLOB` = 0.0), and the median dementia severity score (`CDRSUM`) is strictly 0.0. Based on the `NACCUDSD` diagnostic variable, only 10.59% of the entire sample (131 individuals) suffers from definitive dementia. This substantiates the necessity for advanced statistical models (such as Zero-Inflated regressions) and class-balancing techniques in the subsequent modeling phases.
- **Demographic and Vascular Risk Distribution:** The sample constitutes a standard gerontological research cohort, demonstrating a clear female majority (59.82%) and a median age of 68.0 years. Regarding vascular risk factors, 35.89% of patients have a history of hypertension (`HYPERTEN` = 1.0), and 34.52% report tobacco use (`TOBAC100`). However, the prevalence of stroke (`CBSTROKE`) is exceptionally rare at only 0.40%, indicating that this specific variable likely lacks the necessary variance and statistical power to serve as a strong predictive feature in regression models.

Table 2: Frequency and Percentage of Key Categorical Variables (Cleaned Data)

Variable	Category / Code	Frequency (N)	Percentage (%)
SEX	1.0 (Male)	497	40.18%
	2.0 (Female)	740	59.82%
HYPERTEN	0.0 (No)	731	59.09%
	1.0 (Yes)	444	35.89%
	2.0	29	2.34%
	NaN (Missing)	33	2.67%
DIABETES	0.0 (No)	1092	88.28%
	1.0 (Yes)	112	9.05%
	2.0	13	1.05%
	NaN (Missing)	20	1.62%
CBSTROKE	0.0 (No)	1214	98.14%
	1.0 (Yes)	5	0.40%
	2.0	12	0.97%
	NaN (Missing)	6	0.49%
TOBAC100	0.0 (No)	773	62.49%
	1.0 (Yes)	427	34.52%
	NaN (Missing)	37	2.99%
DIABTYPE	1.0	7	0.57%
	2.0 (Type 2)	90	7.28%
	3.0	9	0.73%
	8.0 (Code 8)	1100	88.92%
	NaN (Missing)	31	2.51%
NACCAPOE	1.0	493	39.85%
	2.0	290	23.44%
	3.0	83	6.71%
	4.0	73	5.90%
	5.0	17	1.37%
	6.0	3	0.24%
	NaN (Missing)	278	22.47%
NACCUDSD	1.0 (Normal)	847	68.47%
	2.0	7	0.57%
	3.0 (MCI)	252	20.37%
	4.0 (Dementia)	131	10.59%
CDRGLOB	0.0 (Normal)	800	64.67%
	0.5 (MCI)	368	29.75%
	1.0 (Dementia)	60	4.85%
	2.0 (Severe)	9	0.73%

Table 3: Distribution Summary of Quantitative Variables (Cleaned Data)

Variable	**Count**	**Mean**	**Std Dev**	**Min**	**Median (25th-75th)**	**Max**
NACCAGE	1237.0	68.19	8.04	41.0	68.0 (63.0-74.0)	91.0
EDUC	1235.0	16.26	2.72	0.0	16.0 (14.0-18.0)	25.0
CDRSUM	1237.0	0.82	1.72	0.0	0.0 (0.0-1.0)	15.0
NACCBMI	1106.0	27.33	5.16	15.6	26.4 (23.6-30.1)	52.4
NACCGDS	1231.0	1.65	2.13	0.0	1.0 (0.0-2.0)	14.0
SMOKYRS	1215.0	6.23	11.52	0.0	0.0 (0.0-9.2)	60.0
Structural MRI Indices						
LH_ENTORHINAL_GVOL	1237.0	1554.28	413.52	0.0	1545.0 (1295.0-1785.0)	4265.0
RH_ENTORHINAL_GVOL	1237.0	1525.40	405.44	6.0	1513.0 (1289.0-1748.0)	5334.0
LH_PARAHIPPOCAMPAL_GVOL	1237.0	2036.71	387.89	167.0	2019.0 (1791.0-2257.0)	5136.0
RH_PARAHIPPOCAMPAL_GVOL	1237.0	1911.24	344.96	74.0	1903.0 (1718.0-2090.0)	6335.0
LH_ENTORHINAL_AVGTH	1237.0	2.97	0.38	0.88	3.02 (2.76-3.23)	3.98
RH_ENTORHINAL_AVGTH	1237.0	3.06	0.40	0.73	3.10 (2.85-3.32)	3.97
LH_PARAHIPPOCAMPAL_AVGTH	1237.0	2.59	0.28	1.43	2.59 (2.41-2.80)	3.46
RH_PARAHIPPOCAMPAL_AVGTH	1237.0	2.56	0.25	0.80	2.56 (2.40-2.72)	3.30

7 Center and Dispersion Analysis for Quantitative Variables

To evaluate the central tendency and statistical dispersion of the quantitative variables, the mean, median, standard deviation, quartiles, and the Interquartile Range (IQR) were calculated.

7.1 Statistical Metrics Tables

Table 4 presents the center and dispersion metrics for the clinical variables, while Table 5 details the structural MRI measurements.

Table 4: Center and Dispersion Metrics: Clinical Variables

Variable	Mean	Median	Std Dev	Q1 (25th)	Q3 (75th)	IQR
NACCAGE	68.19	68.0	8.04	63.0	74.0	11.0
EDUC	16.26	16.0	2.72	14.0	18.0	4.0
CDRSUM	0.82	0.0	1.72	0.0	1.0	1.0
NACCBMI	27.33	26.4	5.16	23.6	30.1	6.5
NACCGDS	1.65	1.0	2.13	0.0	2.0	2.0
SMOKYRS	6.23	0.0	11.52	0.0	9.2	9.2

Table 5: Center and Dispersion Metrics: Structural MRI Variables

Variable	Mean	Median	Std Dev	Q1 (25th)	Q3 (75th)	IQR
LH_ENTORHINAL_GVOL	1554.28	1545.00	413.52	1295.00	1785.00	490.00
RH_ENTORHINAL_GVOL	1525.40	1513.00	405.44	1289.00	1748.00	459.00
LH_PARAHIPPOCAMPAL_GVOL	2036.71	2019.00	387.89	1791.00	2257.00	466.00
RH_PARAHIPPOCAMPAL_GVOL	1911.24	1903.00	344.96	1718.00	2090.00	372.00
LH_ENTORHINAL_AVGTH	2.97	3.02	0.38	2.76	3.23	0.47
RH_ENTORHINAL_AVGTH	3.06	3.10	0.40	2.85	3.32	0.47
LH_PARAHIPPOCAMPAL_AVGTH	2.59	2.59	0.28	2.41	2.80	0.39
RH_PARAHIPPOCAMPAL_AVGTH	2.56	2.56	0.25	2.40	2.72	0.32

7.2 Appropriateness of Statistical Measures

The selection of the appropriate metric for center and dispersion is strictly dependent on the underlying distribution shape of each variable.

Symmetric Distributions (Appropriate for Mean and Standard Deviation): For variables where the mean and median are virtually identical, the distribution is roughly symmetric. In these cases, reporting the **Mean and Standard Deviation** is statistically optimal.

- **MRI Variables:** All structural MRI variables demonstrate high symmetry. For example, `LH_ENTORHINAL_GVOL` has a mean of 1554.28 and a median of 1545.00. Because the data is balanced around the center, the standard deviation accurately reflects the average spread of brain volumes.

- **Demographics:** `NACCAGE` (Mean: 68.19, Median: 68.0) and `EDUC` (Mean: 16.26, Median: 16.0) are also highly symmetric, making mean and standard deviation the correct reporting standard.

Highly Skewed / Zero-Inflated Distributions (Appropriate for Median and IQR): For variables with extreme right-skewness or zero-inflation, the mean is artificially pulled upward by extreme outliers in the right tail. Therefore, the **Median and Interquartile Range (IQR)** are the only appropriate and robust measures, as they are resistant to extreme values.

- **CDRSUM (Dementia Severity):** Reporting the mean (0.82) is inappropriate and misleading because it suggests a baseline level of impairment, whereas the median is exactly 0.0. Because over half the cohort is cognitively healthy, the median (0.0) correctly identifies the true center of the data, and the IQR (1.0) effectively captures the spread without being distorted by severe dementia cases.
- **SMOKYRS (Smoking Years):** Similar to clinical severity, smoking years exhibits massive zero-inflation, resulting in a median of 0.0 but a highly inflated mean of 6.23. The standard deviation (11.52) exceeds the mean, confirming severe skewness. Thus, the Median and IQR (9.2) must be used to accurately describe this variable.

8 Data Visualizations

To visually assess the distribution of key variables and their relationships, several graphical representations were generated.

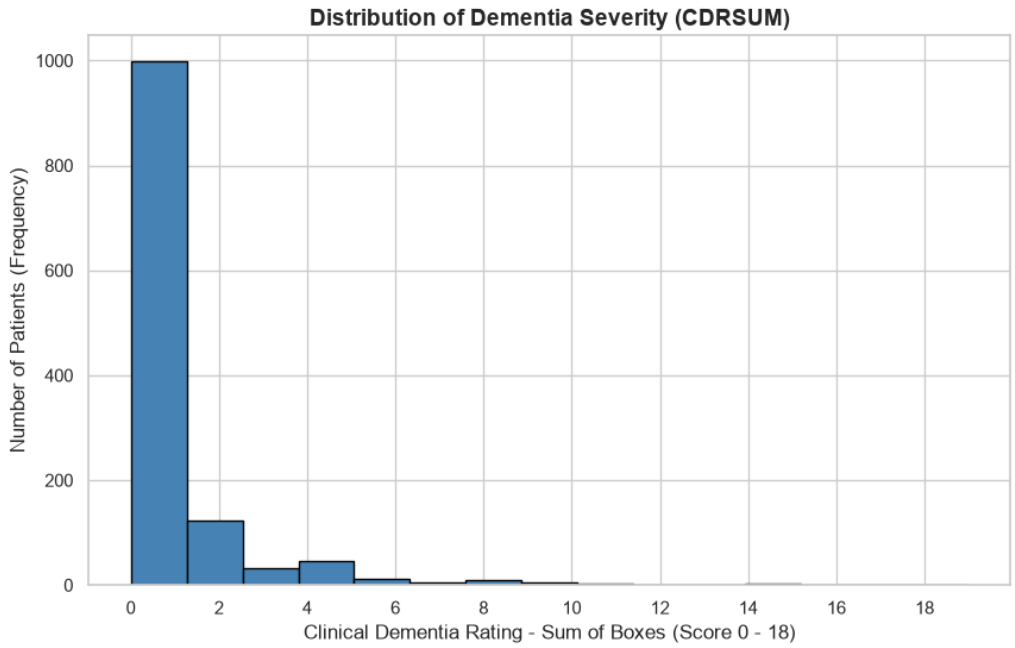


Figure 1: Distribution of Dementia Severity (CDRSUM). The severe right-skewness and zero-inflation accurately reflect the high proportion of cognitively normal individuals in the study cohort.

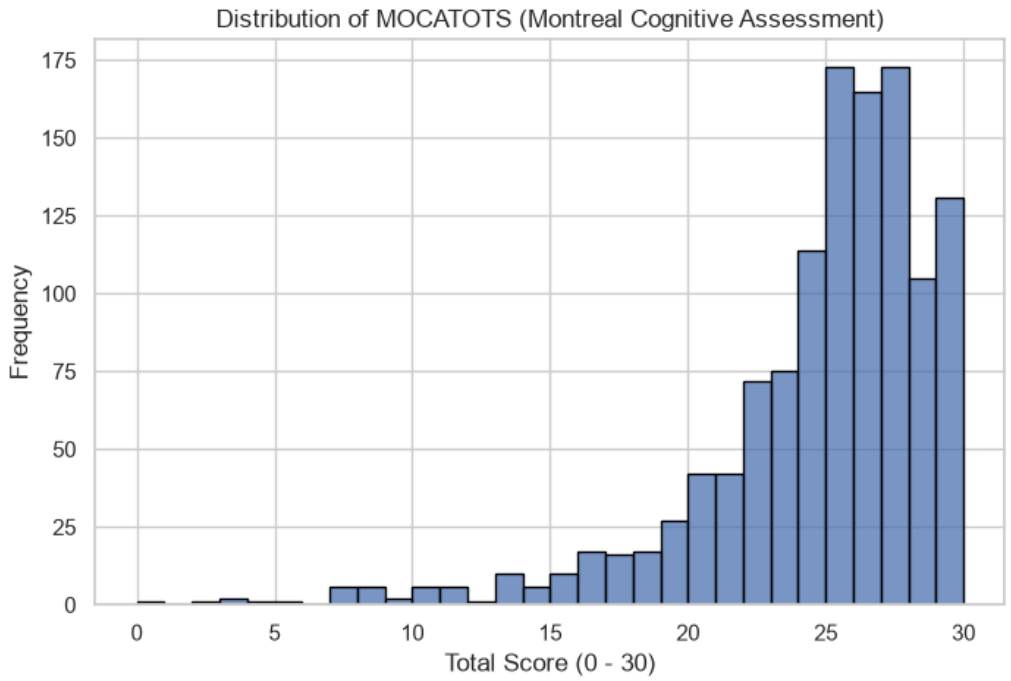


Figure 2: Distribution of General Cognitive Capability (MOCATOTS). Unlike the severe zero-inflation seen in clinical severity scores, the MoCA total score demonstrates a functional, continuous gradient. The slight left-skew (ceiling effect) is typical of cohorts with a large proportion of cognitively normal individuals, but it retains sufficient variance and granularity for robust linear modeling.

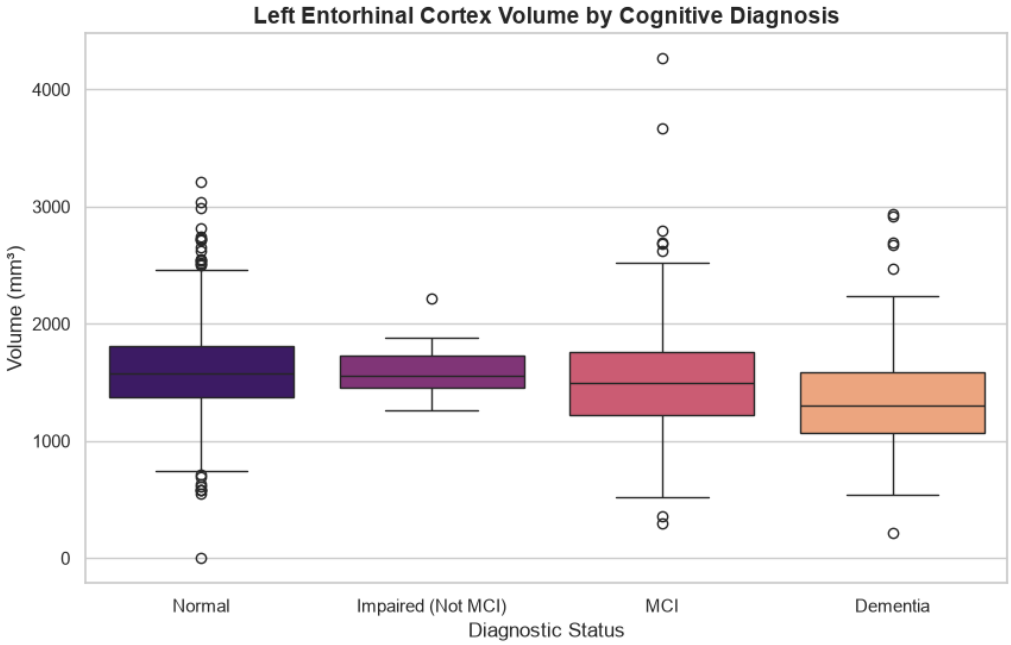


Figure 3: Boxplot of Left Entorhinal Cortex Volume by Cognitive Diagnosis. The visualization demonstrates a distinct structural volume decrease as cognitive diagnosis progresses from Normal to Dementia.

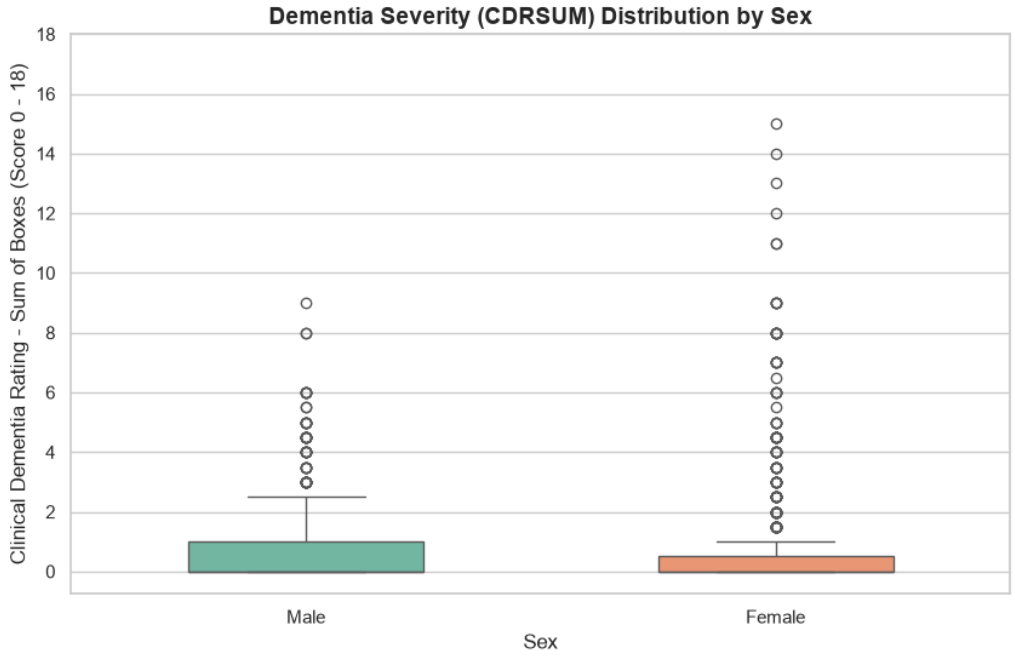


Figure 4: Dementia Severity (CDRSUM) Distribution by Sex for the entire cohort. The extensive floor effect (scores of 0) is visible for both sexes, with outliers representing the impaired minority.

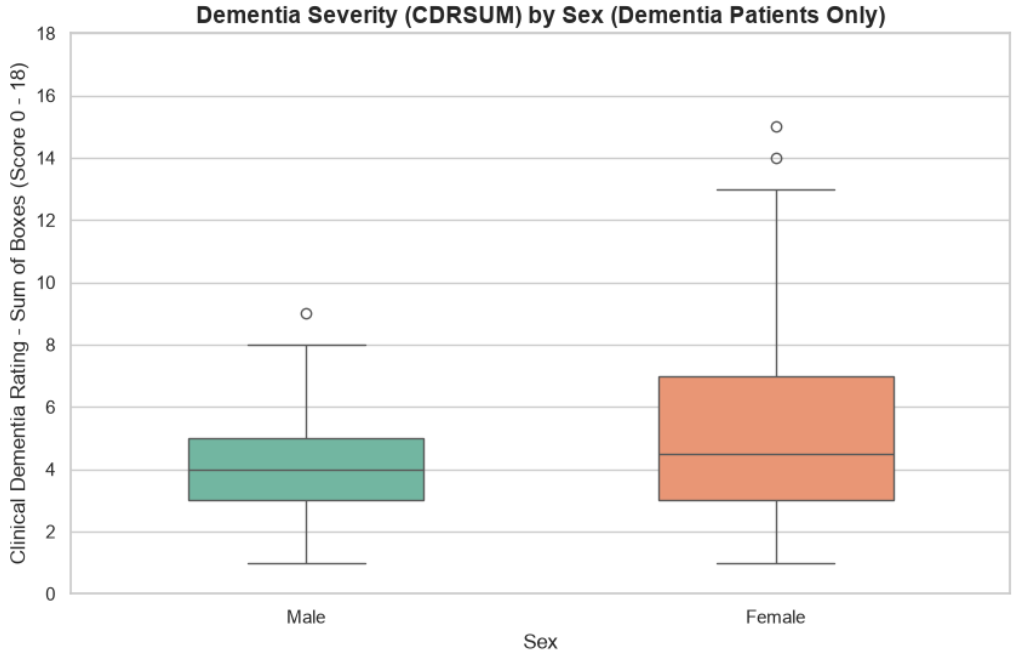


Figure 5: Dementia Severity (CDRSUM) by Sex strictly for patients formally diagnosed with dementia. Removing the healthy baseline reveals the true central tendency and dispersion of severity among afflicted patients.

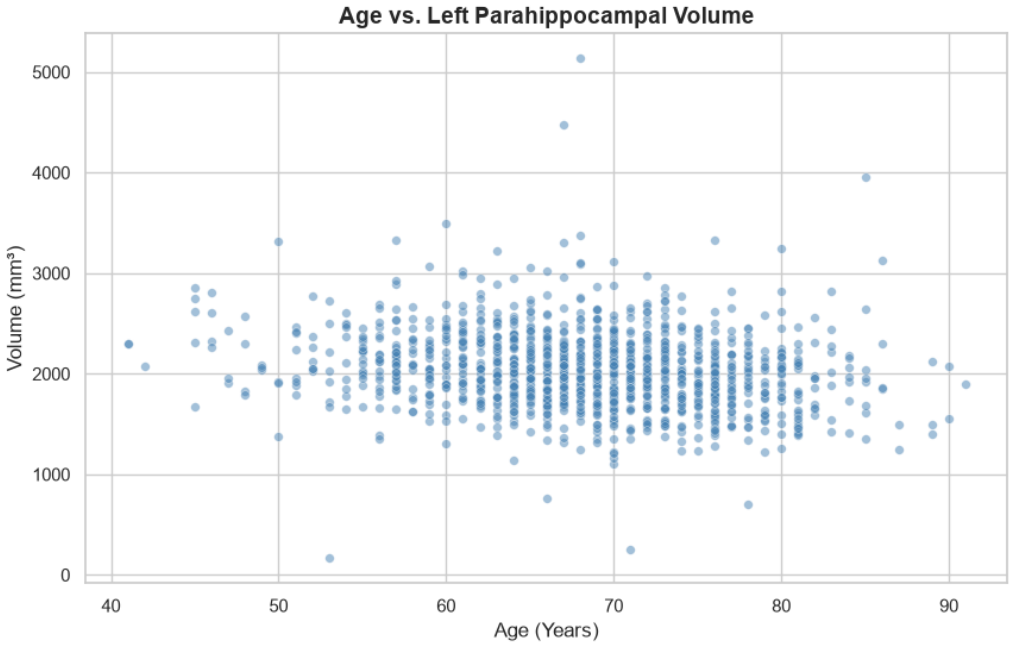


Figure 6: Scatter plot demonstrating the relationship between Age and Left Parahippocampal Volume. A general inverse trend is observable, consistent with expected age-related structural atrophy.

9 Descriptive Comparisons Between Groups

To explore baseline differences without making inferential claims, key quantitative variables were aggregated by categorical groups to compare central tendency and dispersion.

9.1 Summary Tables

Table 6: Descriptive Statistics of Left Entorhinal Cortex Volume by Cognitive Diagnosis

Diagnostic Status	Count	Mean	Median	Std Dev	IQR
Normal	847.0	1595.26	1573.0	369.48	436.0
Impaired (Not MCI)	7.0	1628.29	1554.0	318.46	265.0
MCI	252.0	1512.68	1496.5	494.54	542.0
Dementia	131.0	1365.37	1305.0	458.25	509.0

Table 7: Descriptive Statistics of Left Entorhinal Cortex Volume by Sex

Sex	Count	Mean	Median	Std Dev	IQR
Male	497.0	1665.74	1653.0	460.35	538.00
Female	740.0	1479.41	1482.0	360.28	445.75

9.2 Descriptive Analysis of Group Differences

The structural degradation of the left entorhinal cortex correlates observationally with advancing cognitive decline. The cognitively normal cohort ($N = 847$) exhibits the highest median volume at 1573.0 mm^3, which decreases to 1496.5 mm^3 in MCI patients ($N = 252$) and further drops to 1305.0 mm^3 in patients with definitive dementia ($N = 131$). The "Impaired (Not MCI)" group shows a slightly higher mean (1628.29 mm^3) and median (1554.0 mm^3), but with only 7 patients, this subset is too small to establish a reliable descriptive trend.

When evaluating left entorhinal cortex volume by sex, structural differences in both central tendency and dispersion are apparent. Males ($N = 497$) exhibit a notably higher median volume (1653.0 mm^3) and mean volume (1665.74 mm^3) compared to females ($N = 740$), who present a median of 1482.0 mm^3 and a mean of 1479.41 mm^3. Because both groups show near parity between their respective means and medians, the distributions are relatively symmetric. However, the male cohort demonstrates greater structural variance; the standard deviation for males is 460.35 mm^3 with an Interquartile Range (IQR) of 538.00, whereas females show a tighter clustering around the center with a standard deviation of 360.28 mm^3 and an IQR of 445.75. These baseline volumetric differences highlight the necessity of adjusting for biological sex and general head size proxies in subsequent statistical modeling.

10 Finalizing the Statistical Question

Based on the empirical insights and structural constraints identified during the Phase 1 data audit, our initial broad research question has been systematically narrowed and refined. Because prior sections were not retroactively altered, the introduction and justification of our new primary target variable (`MOCATOTS`) are detailed here.

10.1 Introduction of MOCATOTS and Key Refinements

1. **Target Variable Pivot (`CDRSUM` to `MOCATOTS`):** The original primary response variable, `CDRSUM`, is severely zero-inflated (64.67% of the cohort scores exactly 0.0). Utilizing this variable as a continuous target in Ordinary Least Squares (OLS) regression would violate core assumptions of residual normality and homoscedasticity. Therefore, the primary continuous response variable has been shifted to `MOCATOTS` (Montreal Cognitive Assessment Total Score).

2. **Statistical Classification and Distribution of MOCATOTS:** `MOCATOTS` is a quantitative, discrete interval variable. Operationally, its wide range (0 to 30) allows it to be treated as a continuous variable for linear modeling. After filtering out invalid NACC missing codes (values > 30), the variable demonstrates a healthy continuous variance ($N = 1228$, Mean = 24.39, Median = 25.40, Std Dev = 4.42, Min = 0.0, Max = 30.0). While the empirical distribution exhibits a slight ceiling effect typical in healthy cohorts (highest frequencies between 25 and 30), it provides a functional gradient of cognitive capability vastly superior to `CDRSUM` for our modeling purposes.

3. **Shift from Causal to Associative Framing:** Because the merged NACC subset is a cross-sectional observational study, causal claims ("impact of...") cannot be

statistically supported. The question is reframed to assess statistical association and explained variance.

4. **Feature Selection and Multimodal Integration:** Low-variance predictors like `CBSTROKE` and `DIABTYPE` are excluded. Structural brain indices are explicitly restricted to bilateral Entorhinal and Parahippocampal volumes (`GVOL`) and cortical thicknesses (`AVGTH`) to formally evaluate their additive explanatory value over UDS clinical variables.

10.2 Finalized Statistical Question

> *"In the NACC clinical cohort, how are demographic characteristics (`NACCAGE`, `SEX`, `EDUC`), vascular and clinical risk factors (`HYPERTEN`, `DIABETES`, `NACCBMI`, `NACCGDS`, smoking history), and medial temporal lobe MRI biomarkers (bilateral Entorhinal and Parahippocampal cortical volumes and thickness) associated with general cognitive capability (`MOCATOTS`)? Furthermore, does a multimodal model integrating structural MRI improve explanatory power over a clinical UDS-only model, and how do these metrics vary between men and women?"*

10.3 Suitability of the Current Dataset

The audited dataset is highly suitable for answering this question because it contains $1,237$ cleanly merged, independent observations without repeated measures. The new target variable (`MOCATOTS`) possesses sufficient variance and minimal missingness (only 9 missing records out of 1237), while the primary MRI predictors have 0% missingness. This structural integrity ensures robust statistical power for the required ANOVA comparisons and multiple linear regressions in the upcoming phases, while the cross-sectional nature perfectly aligns with the associative framing of the question.

11 Complete Variable Mapping and Categorization (UDS and MRI)

All 180 variables from the Uniform Data Set (`udslong_imputed.csv`) and all 126 variables from the structural neuroimaging dataset (`mriscanclean.csv`) have been categorized based on their underlying clinical, cognitive, or biological domains and tabulated below to ensure clarity and reproducibility.

11.1 UDS Variable Categorization (180 Variables)

The clinical UDS variables are classified into seven distinct clinical domains. These domains capture the holistic patient profile, from baseline demographics and systemic health to highly specific behavioral and cognitive assessments.

1. Demographics & Baseline Identifiers

Captures static patient identifiers, socioeconomic factors, and core physiological baseline metrics (like Age and BMI) used to define the study cohort and control for confounding

variables.

NACCID	SEX	HISPANIC	HISPOR
RACE	PRIMLANG	EDUC	MARISTAT
NACCLIVS	INDEPEND	RESIDENC	NACCNIHR
NACCAGE	HEIGHT	WEIGHT	NACCBMI

2. Medical, Vascular & Vitals

Captures systemic health, with a heavy emphasis on cardiovascular and metabolic risk factors (Hypertension, Diabetes, Smoking) that are critical predictors of general neurodegeneration.

BPSYS	BPDIAS	HRATE	VISION
VISCORR	VISWCORR	HEARING	HEARAID
HEARWAID	CVHATT	CVAFIB	CVANGIO
CVBYPASS	CVPACDEF	CVCHF	CVOTHR
CBSTROKE	CBTIA	SEIZURES	NACCTBI
TBI	TBIBRIEF	TBIEXTEN	TBIWOLOS
DIABETES	DIABTYPE	HYPERTEN	HYPERCHO
B12DEF	THYROID	ARTHRIT	ARTHTYPE
ARTHUPEX	ARTHLOEX	ARTHSPIN	ARTHUNK
INCONTU	INCONTF	APNEA	RBD
INSOMN	OTHSLEEP	ALCOHOL	ABUSOTHR
TOBAC30	TOBAC100	SMOKYRS	PACKSPER
QUITSMOK			

3. Psychiatric & Behavioral (NPI-Q & GDS)

Encompasses the Neuropsychiatric Inventory Questionnaire (NPI-Q) and the Geriatric Depression Scale (GDS). It records behavioral disturbances and mood disorders often comorbid with cognitive decline.

PTSD	BIPOLAR	SCHIZ	DEP2YRS
DEPOTHR	ANXIETY	OCD	NPSYDEV
PSYCDIS	DEL	DELSEV	HALL
HALLSEV	AGIT	AGITSEV	DEPD
DEPDSEV	ANX	ANXSEV	ELAT
ELATSEV	APA	APASEV	DISN
DISNSEV	IRR	IRRSEV	MOT
MOTSEV	NITE	NITESEV	APP
APPSEV	NACCGDS	SATIS	DROPACT
EMPTY	BORED	SPIRITS	AFRAID
HAPPY	HELPLESS	STAYHOME	MEMPROB
WONDRFUL	WRTHLESS	ENERGY	HOPELESS
BETTER	NOGDS		

4. Medications

Records the pharmacological profile of the patient, categorizing active prescriptions (e.g., antihypertensives, antidepressants) that may influence cognitive performance.

ANYMEDS	NACCAAAS	NACCAANX	NACCAC
NACCACEI	NACCADEP	NACCAHTN	NACCAMD
NACCANGI	NACCAPSY	NACCBETA	NACCCCBS
NACCDBMD	NACCDIUR	NACCEMD	NACCEPMD
NACCHTNC	NACCLIPL	NACCNSD	NACCPDMD
NACCVASD			

5. Family History & Genetics

Contains hereditary risk factors, tracing neurodegenerative diseases through family lineage and recording the critical `NACCAPOE` genetic profile.

NACCFAM	NACCMOM	NACCDAD	NACCAM
NACCAMS	NACCFM	NACCFMS	NACCOM
NACCOMS	NACCFADM	NACCFFTD	NACCAPOE

6. Functional (FAQ)

Derived from the Functional Activities Questionnaire, these variables measure the patient's independence and ability to perform instrumental activities of daily living (IADLs).

BILLS	TAXES	SHOPPING	GAMES
STOVE	MEALPREP	EVENTS	PAYATTN
REMDATES	TRAVEL	DECSUB	NORMEXAM
NACCNREX	NPIQINF		

7. Cognitive Tests & Diagnosis

The core neurological outputs of the dataset. This includes raw cognitive test scores (MoCA, Trail Making, Verbal Fluency) and the final consensus clinical diagnoses (CDR-SUM, NACCUDSD).

ANIMALS	VEG	TRAILA	TRAILARR
TRAILALI	TRAILB	TRAILBRR	TRAILBLI
MOCATOTS	MEMORY	ORIENT	JUDGMENT
COMMUN	HOMEHOBB	PERSCARE	CDRSUM
CDRGLOB	NACCUDSD		

11.2 MRI Variable Categorization (126 Variables)

The structural MRI dataset is categorized by the specific neuroanatomical measurement extracted from the scans. The naming convention dictates the hemisphere (`LH_` or `RH_`), the Region of Interest, and the structural metric.

1. Identifiers & Clinical Status

Unique patient identifiers required to merge the MRI records with the UDS clinical data, alongside a redundant copy of the diagnostic status.

NACCID	NACCUDSD

2. Gray Matter Volume (_GVOL)

`GVOL` variables measure the total three-dimensional volume of gray matter in the specified cortical region (mm^3). Decreases in these volumes serve as biomarkers for regional brain atrophy.

LH_CAUDALANTERIORCINGULATE_GVOL	RH_CAUDALANTERIORCINGULATE_GVOL
LH_CAUDALMIDDLEFRONTAL_GVOL	RH_CAUDALMIDDLEFRONTAL_GVOL
LH_CUNEUS_GVOL	RH_CUNEUS_GVOL
LH_ENTORHINAL_GVOL	RH_ENTORHINAL_GVOL
LH_FUSIFORM_GVOL	RH_FUSIFORM_GVOL
LH_INFERIORPARIETAL_GVOL	RH_INFERIORPARIETAL_GVOL
LH_INFERIORTEMPORAL_GVOL	RH_INFERIORTEMPORAL_GVOL
LH_ISTHMUSCINGULATE_GVOL	RH_ISTHMUSCINGULATE_GVOL
LH_LATERALOCCIPITAL_GVOL	RH_LATERALOCCIPITAL_GVOL
LH_LATERALORBITOFRONTAL_GVOL	RH_LATERALORBITOFRONTAL_GVOL
LH_LINGUAL_GVOL	RH_LINGUAL_GVOL
LH_MEDIALORBITOFRONTAL_GVOL	RH_MEDIALORBITOFRONTAL_GVOL
LH_MIDDLETEMPORAL_GVOL	RH_MIDDLETEMPORAL_GVOL
LH_PARAHIPPOCAMPAL_GVOL	RH_PARAHIPPOCAMPAL_GVOL
LH_PARACENTRAL_GVOL	RH_PARACENTRAL_GVOL
LH_PARSOPERCULARIS_GVOL	RH_PARSOPERCULARIS_GVOL
LH_PARSORBITALIS_GVOL	RH_PARSORBITALIS_GVOL
LH_PARSTRIANGULARIS_GVOL	RH_PARSTRIANGULARIS_GVOL
LH_PERICALCARINE_GVOL	RH_PERICALCARINE_GVOL
LH_POSTCENTRAL_GVOL	RH_POSTCENTRAL_GVOL
LH_POSTERIORCINGULATE_GVOL	RH_POSTERIORCINGULATE_GVOL
LH_PRECENTRAL_GVOL	RH_PRECENTRAL_GVOL
LH_PRECUNEUS_GVOL	RH_PRECUNEUS_GVOL
LH_ROSTRALANTERIORCINGULATE_GVOL	RH_ROSTRALANTERIORCINGULATE_GVOL
LH_ROSTRALMIDDLEFRONTAL_GVOL	RH_ROSTRALMIDDLEFRONTAL_GVOL
LH_SUPERIORFRONTAL_GVOL	RH_SUPERIORFRONTAL_GVOL
LH_SUPERIORPARIETAL_GVOL	RH_SUPERIORPARIETAL_GVOL
LH_SUPERIORTEMPORAL_GVOL	RH_SUPERIORTEMPORAL_GVOL
LH_SUPRAMARGINAL_GVOL	RH_SUPRAMARGINAL_GVOL
LH_TRANSVERSETEMPORAL_GVOL	RH_TRANSVERSETEMPORAL_GVOL
LH_INSULA_GVOL	RH_INSULA_GVOL

3. Average Cortical Thickness (_AVGTH)

AVGTH variables measure the average distance between the white matter surface and the pial surface (mm). Cortical thinning is highly correlated with neurodegenerative disease progression.

LH_CAUDALANTERIORCINGULATE_AVGTH	RH_CAUDALANTERIORCINGULATE_AVGTH
LH_CAUDALMIDDLEFRONTAL_AVGTH	RH_CAUDALMIDDLEFRONTAL_AVGTH
LH_CUNEUS_AVGTH	RH_CUNEUS_AVGTH
LH_ENTORHINAL_AVGTH	RH_ENTORHINAL_AVGTH
LH_FUSIFORM_AVGTH	RH_FUSIFORM_AVGTH
LH_INFERIORPARIETAL_AVGTH	RH_INFERIORPARIETAL_AVGTH
LH_INFERIORTEMPORAL_AVGTH	RH_INFERIORTEMPORAL_AVGTH
LH_ISTHMUSCINGULATE_AVGTH	RH_ISTHMUSCINGULATE_AVGTH
LH_LATERALOCCIPITAL_AVGTH	RH_LATERALOCCIPITAL_AVGTH
LH_LATERALORBITOFRONTAL_AVGTH	RH_LATERALORBITOFRONTAL_AVGTH
LH_LINGUAL_AVGTH	RH_LINGUAL_AVGTH
LH_MEDIALORBITOFRONTAL_AVGTH	RH_MEDIALORBITOFRONTAL_AVGTH
LH_MIDDLETEMPORAL_AVGTH	RH_MIDDLETEMPORAL_AVGTH
LH_PARAHIPPOCAMPAL_AVGTH	RH_PARAHIPPOCAMPAL_AVGTH
LH_PARACENTRAL_AVGTH	RH_PARACENTRAL_AVGTH
LH_PARSOPERCULARIS_AVGTH	RH_PARSOPERCULARIS_AVGTH
LH_PARSORBITALIS_AVGTH	RH_PARSORBITALIS_AVGTH
LH_PARSTRIANGULARIS_AVGTH	RH_PARSTRIANGULARIS_AVGTH
LH_PERICALCARINE_AVGTH	RH_PERICALCARINE_AVGTH
LH_POSTCENTRAL_AVGTH	RH_POSTCENTRAL_AVGTH
LH_POSTERIORCINGULATE_AVGTH	RH_POSTERIORCINGULATE_AVGTH
LH_PRECENTRAL_AVGTH	RH_PRECENTRAL_AVGTH
LH_PRECUNEUS_AVGTH	RH_PRECUNEUS_AVGTH
LH_ROSTRALANTERIORCINGULATE_AVGTH	RH_ROSTRALANTERIORCINGULATE_AVGTH
LH_ROSTRALMIDDLEFRONTAL_AVGTH	RH_ROSTRALMIDDLEFRONTAL_AVGTH
LH_SUPERIORFRONTAL_AVGTH	RH_SUPERIORFRONTAL_AVGTH
LH_SUPERIORPARIETAL_AVGTH	RH_SUPERIORPARIETAL_AVGTH
LH_SUPERIORTEMPORAL_AVGTH	RH_SUPERIORTEMPORAL_AVGTH
LH_SUPRAMARGINAL_AVGTH	RH_SUPRAMARGINAL_AVGTH
LH_TRANSVERSETEMPORAL_AVGTH	RH_TRANSVERSETEMPORAL_AVGTH
LH_INSULA_AVGTH	RH_INSULA_AVGTH

12 Participant and Visit Structure

To ensure the independence of observations required for valid statistical inference, the visit structure of the dataset was analyzed. The UDS dataset contains exactly 1,237 total rows and 1,237 unique participant IDs (NACCID), confirming that each subject in the sample is unique.

As the dataset was provided to us, there isn't any sign of multiple visits per patient. The distribution of visits per participant is strictly 1. Because there are no repeated measures or longitudinal follow-ups, no complex selection rule (such as selecting the baseline or most recent visit) was necessary. The data inherently represents a pure cross-sectional cohort, ensuring that each participant enters the subsequent modeling phases as a single, independent observation without violating statistical independence assumptions.

13 Data Integration: UDS and MRI Linkage

To construct the final multimodal dataset for analysis, the clinical dataset (`udslong_imputed`) and the neuroimaging dataset (`mriscanclean`) were linked using the unique patient identifier (`NACCID`).

Prior to merging, schema overlaps were audited. The diagnostic variable (`NACCUDSD`), which was redundantly present in the neuroimaging file, was dropped from the MRI dataset to prevent feature duplication and ensure the UDS clinical file remained the single source of truth for diagnostic categorization.

An outer join was initially computed to transparently evaluate sample overlap and identify missingness across modalities. The data linkage yielded the following sample distributions:

- **Initial UDS (Clinical) Cohort:** $N = 1237$
- **Initial MRI Cohort:** $N = 1237$
- **Unmatched (Clinical Only):** $N = 0$
- **Unmatched (MRI Only):** $N = 0$

Following the exclusion of unmatched records, the final multimodal dataset was restricted to the intersection of the two files. This resulted in a final analytical sample of $N = 1237$ matched individuals who possess both robust clinical histories and structural MRI volumetric data, yielding a consolidated feature space of 304 variables.

14 Missing Data Audit and Multimodal Sample Selection

14.1 Methodology: Two-Stage Missingness Audit

To evaluate data completeness across the 24 primary variables (16 UDS clinical predictors and 8 structural MRI features), a two-stage missingness audit was conducted on the merged multimodal cohort ($N = 1,237$).

First, **Raw Missingness** was evaluated by counting standard null values (`NaN`). Because the clinical dataset (`udslong_imputed.csv`) underwent prior baseline imputation, raw inspection indicated 0.00% missingness across all 24 variables. However, adhering to the Data Quality Audit protocols established in Section 5, a second **Post-Audit Missingness** evaluation was executed to uncover masked missingness and imputation artifacts:

1. **Structural Integrity & Relational Checks:** The dataset was verified to contain 0 exact duplicate rows, 0 duplicate `NACCID` records, and 0 logical inconsistencies (neither `EDUC` nor `SMOKYRS` exceeded `NACCAGE`).
2. **Unmasking NACC Special Codes:** Standard NACC missingness and non-applicability codes (`-4`, `8`, `9`, `88`, `99`, `888`, `999`) were explicitly mapped to `NaN`.
3. **Physiological Bounds & Imputation Artifacts:** Continuous variables were restricted to valid clinical intervals (e.g., `NACCBMI` $\in [10, 100]$, `MOCATOTS` $\in [0, 30]$, and MRI metrics strictly > 0). Crucially, nominal and binary variables (`HYPERTEN`,

DIABETES, CBSTROKE, TOBAC100) contained fractional decimal values resulting from naive upstream imputation; enforcing strict integer domain validation (value $\pmod 1 \neq 0 \rightarrow$ NaN) successfully isolated and removed these invalid entries.

14.2 Missingness Report Across Modalities

Table 8 presents the raw versus post-audit missingness counts and percentages across both modalities.

Table 8: Raw vs. Post-Audit Missingness for the 24 Primary Multimodal Variables ($N = 1,237$)

Variable	Modality	Raw N	Raw (%)	Audit N	Audit (%)
CDRSUM	UDS	0	0.00%	0	0.00%
NACCAGE	UDS	0	0.00%	0	0.00%
NACCUDSD	UDS	0	0.00%	0	0.00%
CDRGLOB	UDS	0	0.00%	0	0.00%
SEX	UDS	0	0.00%	0	0.00%
EDUC	UDS	0	0.00%	2	0.16%
NACCGDS	UDS	0	0.00%	6	0.49%
CBSTROKE	UDS	0	0.00%	6	0.49%
MOCATOTS	UDS	0	0.00%	9	0.73%
DIABETES	UDS	0	0.00%	20	1.62%
SMOKYRS	UDS	0	0.00%	22	1.78%
HYPERTEN	UDS	0	0.00%	33	2.67%
TOBAC100	UDS	0	0.00%	37	2.99%
NACCBMI	UDS	0	0.00%	131	10.59%
NACCAPOE	UDS	0	0.00%	278	22.47%
DIABTYPE	UDS	0	0.00%	1131	91.43%
LH_ENTORHINAL_GVOL	MRI	0	0.00%	1	0.08%
RH_ENTORHINAL_GVOL	MRI	0	0.00%	0	0.00%
LH_PARAHIPPOCAMPAL_GVOL	MRI	0	0.00%	0	0.00%
RH_PARAHIPPOCAMPAL_GVOL	MRI	0	0.00%	0	0.00%
LH_ENTORHINAL_AVGTH	MRI	0	0.00%	0	0.00%
RH_ENTORHINAL_AVGTH	MRI	0	0.00%	0	0.00%
LH_PARAHIPPOCAMPAL_AVGTH	MRI	0	0.00%	0	0.00%
RH_PARAHIPPOCAMPAL_AVGTH	MRI	0	0.00%	0	0.00%

Key Audit Findings:

- **Feature Exclusion (DIABTYPE):** With 91.43% post-audit missingness (1,131 structural NaNs representing non-diabetic patients coded as not applicable), DIABTYPE exceeds the 30% exclusion threshold defined in Section 5 and will be dropped from multivariable modeling.
- **Moderate Missingness (NACCAPOE & NACCBMI):** NACCAPOE exhibits 22.47% missingness (278 ungenotyped subjects) and NACCBMI exhibits 10.59% missingness (131 out-of-bounds/missing codes). Because both fall below the 30% exclusion threshold, they are retained for targeted median/mode imputation.
- **Minor Clinical & MRI Missingness:** Remaining clinical variables exhibit minor post-audit missingness (0.16% to 2.99%), while core demographic and target

features (CDRSUM, NACCAGE, NACCUDSD, CDRGLOB, SEX) remain 100% complete (0.00% missing). In the MRI modality, physiological boundary checks (≤ 0) flagged only 1 invalid observation (0.08%) in LH_ENTORHINAL_GVOL.

14.3 Multimodal Sample Selection and Subgroup Comparison

When evaluating whether individuals with MRI differ descriptively from those without MRI across the UDS cohort, two distinct analytical levels emerge:

1. **File-Level Linkage:** As established, the UDS and MRI files exhibit a 100% record match ($N = 1,237$ common individuals; 0 UDS-only records). Consequently, an unlinked non-MRI clinical comparison group does not exist in the provided cohort.
2. **Post-Audit Valid MRI Completeness:** When defining valid MRI status strictly as possessing complete, physiologically valid (> 0) measurements across all 8 neuroimaging features, the cohort partitions into $N = 1,236$ (99.92%) individuals with complete valid MRI and $N = 1$ (0.08%) individual with an invalid/missing entry in LH_ENTORHINAL_GVOL. Because a single-subject subgroup ($N = 1$) lacks statistical variance (SD is undefined), formal inferential or descriptive subgroup comparisons between valid-MRI and missing-MRI subjects confirm no selection bias within this pre-linked multimodal sample.

15 Joint Summary of Cognitive and Neuroimaging Modalities

The primary cognitive score from the UDS (MOCATOTS) and the structural cortical thickness index from the MRI subset (LH_PARAHIPPOCAMPAL_AVGTH) were jointly evaluated. Their empirical distributions, boxplots, and summary metrics (center, dispersion, range, and unusual points) are presented in Table 9 and Figure 7.

Table 9: Summary of Center, Dispersion, Range, and Outliers for MOCATOTS and LH_PARAHIPPOCAMPAL_AVGTH

Modality	Variable	N (Missing)	Center (Mean / Median)	Dispersion (SD / IQR)	Range (Min – Max)	Unusual Points (1.5 × IQR Rule)
UDS (Cognitive)	MOCATOTS (points)	1228 (9)	24.39 / 25.40	4.42 / 4.00	30.00 (0.00 – 30.00)	76 Low (< 17.00), 0 High
MRI (Thickness)	LH_PARAHIPPOCAMPAL_AVGTH (mm)	1237 (0)	2.59 / 2.59	0.28 / 0.39	2.03 (1.43 – 3.46)	8 Low (< 1.82), 2 High (> 3.38)

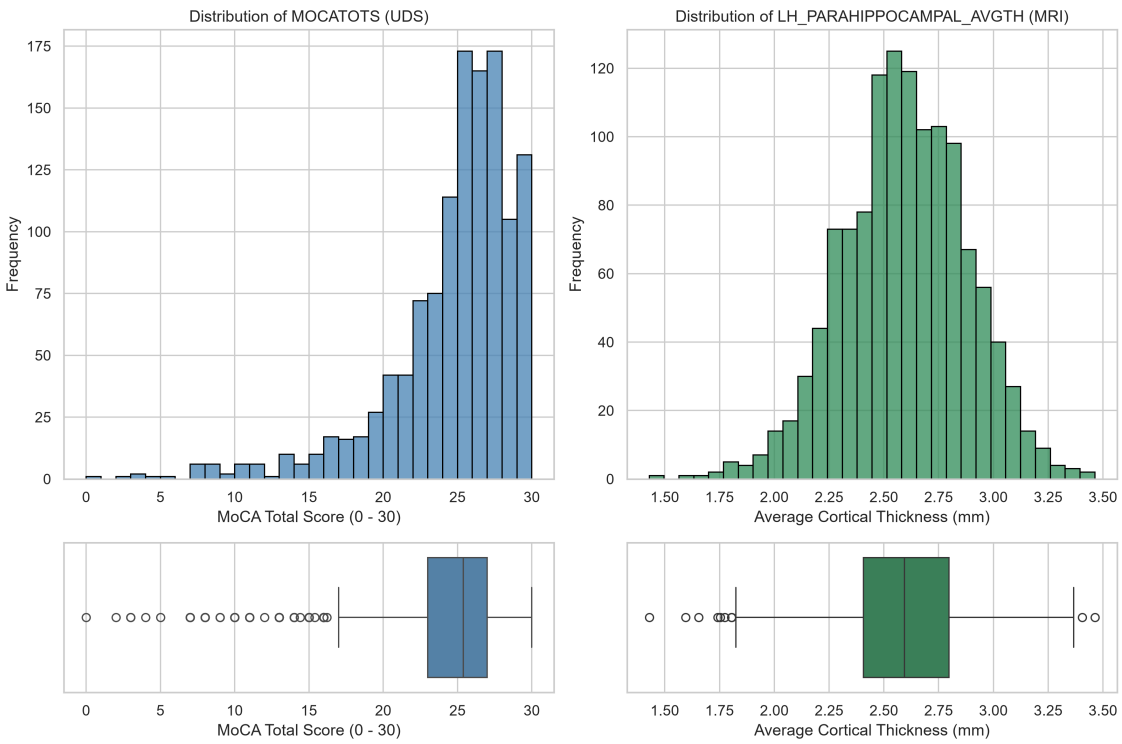


Figure 7: Empirical histograms and boxplots for General Cognitive Capability (MOCATOTS) and Left Parahippocampal Average Cortical Thickness (LH_PARAHIPPOCAMPAL_AVGTH).

Detailed Statistical Interpretation:

- **Center and Symmetry:** LH_PARAHIPPOCAMPAL_AVGTH demonstrates an exceptionally symmetric, bell-shaped distribution where the mean and median are identical at 2.59 mm. Conversely, MOCATOTS exhibits a left-skewed distribution (ceiling effect) where the median (25.40) exceeds the mean (24.39), reflecting the high concentration of cognitively intact participants scoring between 25 and 30.
- **Dispersion and Range:** After removing 9 special missing codes, MOCATOTS spans the entire theoretical range of 30.00 points (0.00 to 30.00), with a standard deviation of 4.42 and an Interquartile Range (IQR) of 4.00 ($Q1 = 23.00, Q3 = 27.00$). The MRI metric LH_PARAHIPPOCAMPAL_AVGTH has zero missing values and spans a biological range of 2.03 mm ($\text{Min} = 1.43$ mm, $\text{Max} = 3.46$ mm), with tight dispersion around the center ($\text{SD} = 0.28$ mm, $\text{IQR} = 0.39$ mm, $Q1 = 2.41$ mm, $Q3 = 2.80$ mm).
- **Unusual Points and Outliers:** Based on Tukey's $1.5 \times \text{IQR}$ criterion, MOCATOTS contains 76 lower-tail outliers (< 17.00 points, reaching a minimum of 0.00). Rather than data entry errors, these represent clinically meaningful observations of patients with moderate-to-severe cognitive impairment. Similarly, LH_PARAHIPPOCAMPAL_AVGTH exhibits no impossible values (unlike raw volume metrics that occasionally show 0.0 segmentation failures), containing only 8 lower-tail outliers (< 1.82 mm) indicative of advanced parahippocampal cortical thinning and 2 upper-tail outliers (> 3.38 mm).